then we find that,

$$\begin{aligned}\tilde{H}(Q,P) &= \frac{P^2}{2} + \frac{h\Omega^2 PQ}{2} + \frac{\Omega^2 Q^2}{2} \\ &= \frac{1}{2}p^2 - \frac{h\Omega^2}{2}pQ + \frac{\Omega^2 Q^2}{2} \\ &= \tilde{H}(q,p)\end{aligned}$$

This means that $\tilde{H}$ is a conserved quantity of the numerical method.

Recall that the graph of

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

is an ellipse with major and minor axes aligned to the coordinate axes. If ε is a small value, then

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \varepsilon xy = 1$$

will be a slightly rotated ellipse (with slightly different major and minor axes). Thus we can think of the energy surface of the numerical method as being a small perturbation of the ellipse which represents the 'energy surface' (energy curve, in this case) of the harmonic oscillator itself.

For some values of the stepsize, it is possible that the points generated by the numerical method will constitute a finite periodic sequence. For example let $\Omega = 1$ and use the large stepsize of $h = 1$. Then from the initial point $(q_0, p_0) = (1, 0)$ we have $q_1 = q_0 + p_0 = 1$, and $p_1 = p_0 - q_1 = -1$, so we visit $(1, -1)$, then $(0, -1)$, then $(-1, 0)$, $(-1, 1)$, $(0, 1)$ and finally return to $(1, 0)$ (diagrammed in the left panel of Fig. 3.1). For a different choice of the stepsize, the point sequence may

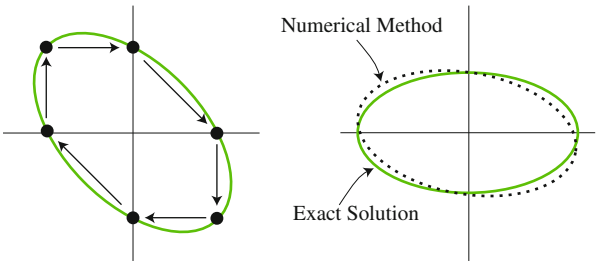


Fig. 3.1 Numerical solutions of the harmonic oscillator may cycle through a finite number of discrete points (*left*) or may gradually fill in an ellipse that is just a slight perturbation of the correct energy level set (*right*)